Why Protection?

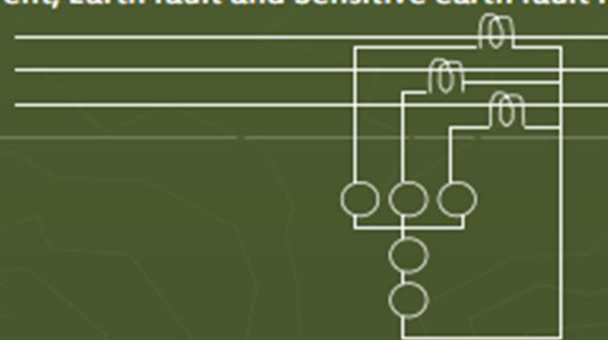
- Any equipment is subjected to faults both from internal as well as external problems – Component failures, Insulation failures, Lightning strokes, Tree fouling, Bird fouling, Animal electrocution, Abnormal operating conditions, Manual errors, etc.
- Thermal damage
- Need to be protected to avoid damages
- And also to minimize outages from system stability point of view
- Extent of damage depends on both amplitude and duration
- It is essential to know the severity and boundary values
- Voltage and current distribution in a network, due to fault must be evaluated
- Fault calculations will reveal the gravity

415 V LT Distribution lines

- Protected by Fuses only
- The simplest form of protection
- Sacrificing type
- Operate on actual quantities

HT Distribution lines (11 kV to 33 kV)

- Actual quantities cannot be handled straightaway – Operate on replica quantities – Instrument transformers needed
- Protection by employing relays and switchgear
- Over current, Earth fault and Sensitive earth fault relays are used



DC Scheme used to operate the trip coil

- O/C relay setting – Ampacity & Line end fault current are the deciding factors
- E/F relay – Above the expected unbalance
- SEF relay – Low pick up setting – High time setting – Applicable only for highly balanced lines

WHAT IS SWITCHGEAR?

A device or a combination of devices, primarily intended for the purpose of making, carrying and breaking electric currents in circuits under normal conditions as well as under abnormal (faulty) conditions.

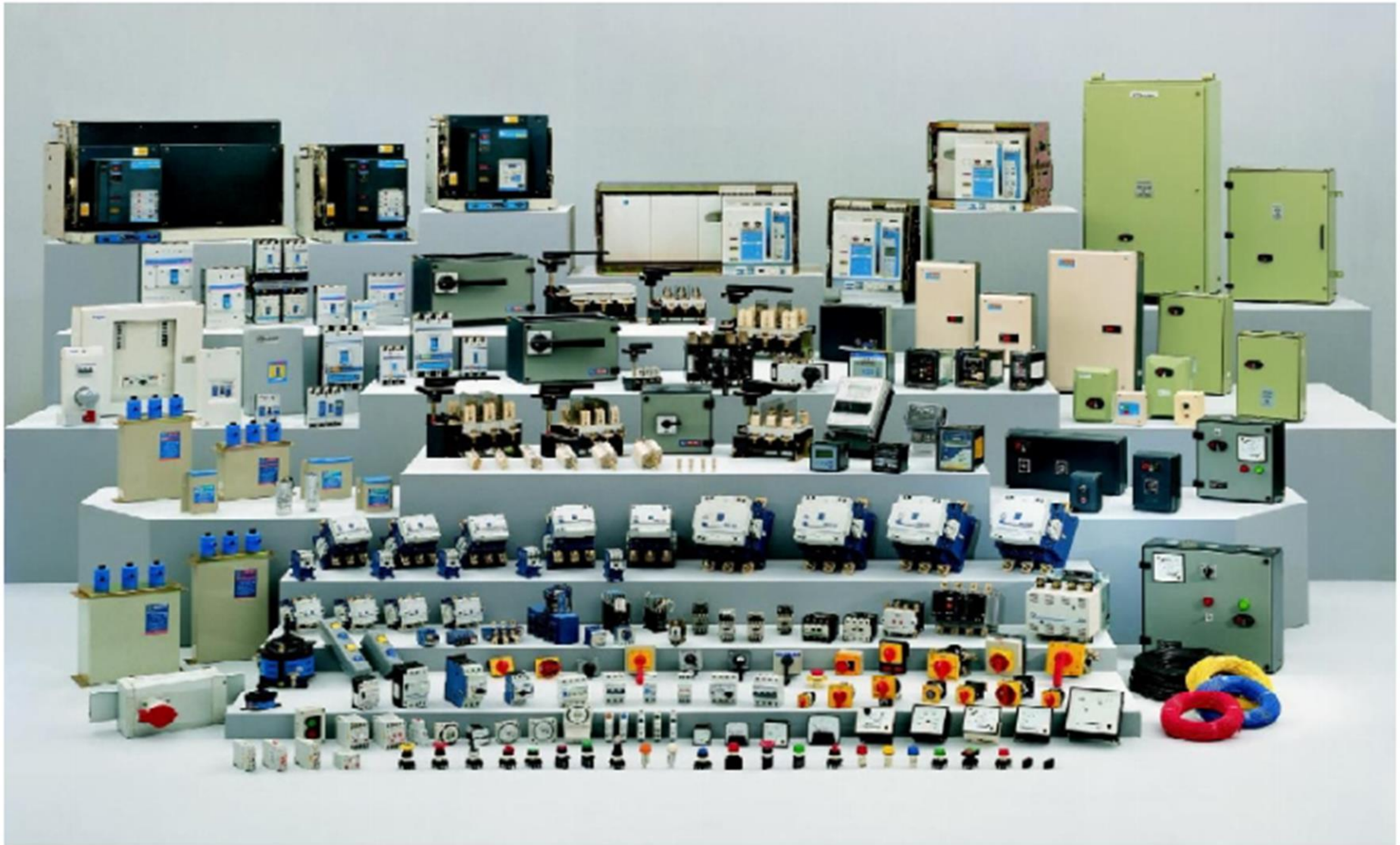
EXAMPLES OF LV SWITCHGEAR

- Air Circuit Breakers (ACBs)
- Moulded Case Circuit Breakers (MCCBs)
- Motor Protection Circuit Breakers (MPCBs)
- Miniature Circuit Breakers (MCBs)
- Earth Leakage Circuit Breakers (ELCBs)
- Residual Current Circuit Breakers (RCCBs)
- Air Break Contactors
- Vacuum Contactors
- Thermal Overload Relays

EXAMPLES OF LV SWITCHGEAR

- Switches/Disconnectors/ Switch-Disconnectors
- HRC Fuses
- Switch Fuse Units (SFU)
- Fuse Switch Units (FSU)
- Switch – Disconnector - Fuse (SDF) Units
- Current Transformers
- Voltage Transformers
- Selector Switches
- Push Buttons

Range of Low Voltage Switchgear



NEED OF PROTECTIVE SYSTEMS

- An EPS consists of Generators, Transformers, Transmission line and Distribution lines, etc..
- SC and other abnormal conditions often occur in Power systems.
- A Protective scheme include CB and Protective relays to isolate the faulty section from the healthy.
- A CB can disconnect the faulty element, when it is called upon to do so by the relay.

RELAY

- It is a device which sense the abnormal conditions on a PS by constantly monitoring electrical quantities.- voltage, current, phase angle(direction) , frequency
- The function of the relay is to detect and locate a fault and issue a command to the CB to disconnect the faulty element.
- A Protective relay does not anticipate or prevent the occurrence of a fault, rather it takes action only after fault has occurred.
- However – Buchholz relay is an exception.

OTHER ABNORMAL CONDITION

- Over speed of generators and motors
- Over voltage
- Under frequency
- Loss of excitation
- Overheating of excitation of stator and rotor

- Note: Cost of protective scheme should be about 5% of the total cost of the systems.

NATURE AND CAUSES OF FAULTS

- Insulation failure - SC
- Conducting path failure
- Over voltage due to lightning/ switching surge
- External conducting objects falling on overhead lines
- Fine dust, cement accumulation – on pin insulator – reduces insulation strength – causes flash over
- Birds body touches line and earth wire
- Joint failure in cable and overhead line
- Unbalanced current – rotating machine – set up harmonic current – heating

NATURE AND CAUSES OF FAULTS

- Other causes are (overhead lines): direct lighting strokes, aircraft, snakes, ice & snow loading, storms, earth quakes
- Other causes(Gen, Transformer): ageing, accidental operation, wrong connection, defect in the protective devices, sometime poor quality of system components, faulty system design .

TYPES OF FAULT

- Symmetrical fault : 3 phase fault (with / without the Ground)- to determine the system fault level
- Unsymmetrical fault:

- LG – due to failure of insulation, due to phase conductor breaking and falling to the ground
- L-L-G –
- L – L

- Open Circuited faults
- Winding fault

Simultaneous fault

- Two or more faults occurring simultaneously

  Same/ different type – at same or different point

cross – country earth fault

- The simultaneous presence of an L-G fault at one point and a second L-G fault on another phase at some other point is an example of two faults of the same type at two different points. If these two L-G faults are on the same section of the line, they are treated as L-L-G fault. If they occur in different line sections, it is known as a cross – country earth fault.

Cross-country faults

- Cross-country faults are common on systems grounded through high impedance or Peterson coil but they are rare in a solidly grounded systems.

BASIC RELAY TERMINOLOGY

Relay: A relay is an automatic device by means of which an electrical circuit is indirectly controlled (opened and closed) and is governed by a change in the same or another electrical circuit.

Protective Relay

A Protecting relay is an automatic device which detects an abnormal condition in an electrical circuit and causes a circuit breaker to isolate the faulty element of the systems. In some cases it may give an alarm or visible indication to alert operator

Operating Force or torque

A force or torque which tends to close the contacts of the relay

Restraining force or torque

A force or torque which opposes the operating force/torque

Actuating quantity

An electrical quantity (current, voltage, etc.) to which a relay responds

Pick-up(level)

The threshold value of the actuating quantity (current , voltage..) above which the relay operates

Reset on drop-out

The threshold value of the actuating quantity below which the relay is de-energised and returns to normal position or state

Operating time

It is the time which elapses from the instant at which the actuating quantity exceeds the relays pick-up value to the instant at which the relay closes its contacts.

Reset time

It is the time which elapses from the moment the actuating quantity falls below its reset value to the instant when the relay comes back to its normal position

Setting

The value of the actuating quantity at which the relay is set to operate

Primary protection

If a fault occurs, it is the duty of the primary protective scheme to clear the fault. It acts as a first line of defence. If it fails, the back-up protection clears the fault

Back up protection

The back-up protection is designed to clear the fault if the primary protection fails. It acts as a second line of defence.

Back-up relay

A back-up relay operates after a slight delay, if the main relay fails to operate.

Instantaneous relay

An instantaneous relay has no intentional time delay in its operation.

Inverse time relay

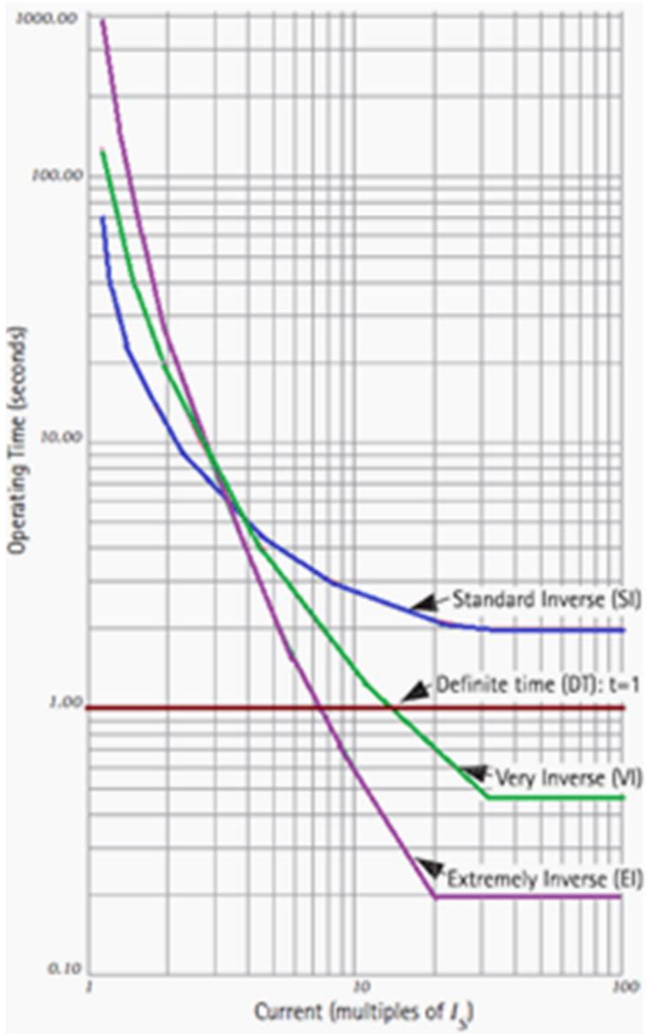
A relay in which the operating time is inversely proportional to the magnitude of the operating current.

Definite time relay

The relay in which the operating time is independent of the magnitude of the actuating current.

Inverse definite minimum time (IDMT) relay

A relay which gives an inverse time characteristics at lower values of the operating current and definite time characteristics at higher values of the operating current.


1000.00
100.00
10.00
1.00
0.10
Operating Time (seconds)
1
10
100
Current (multiples of I_s)
Standard Inverse (SI)
Definite time (DT): t=1
Very Inverse (VI)
Extremely Inverse (EI)

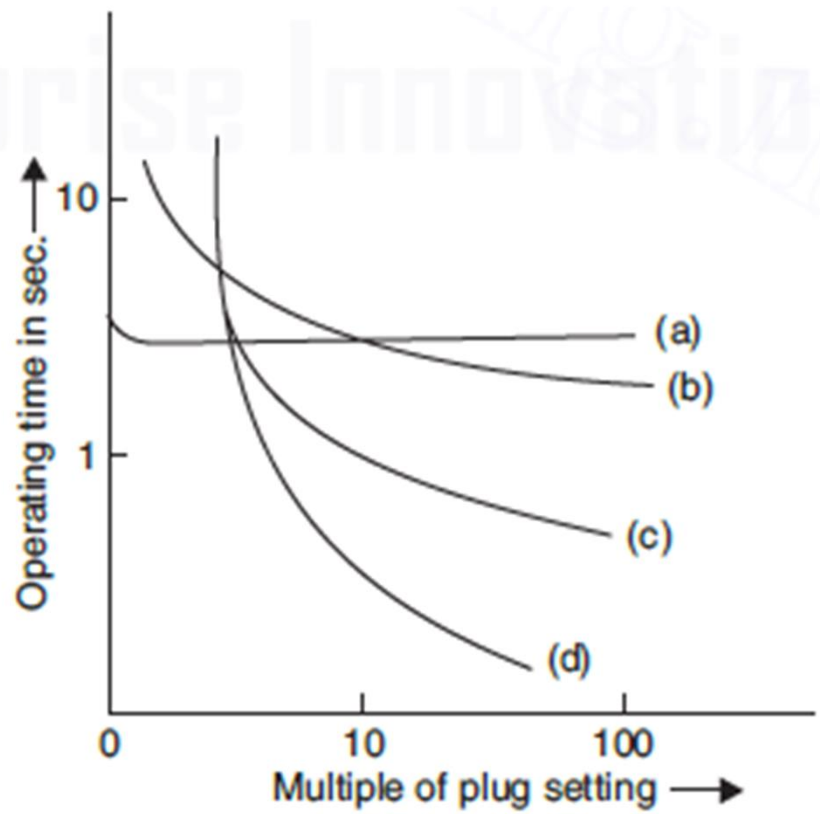


Fig. 14.6 Characteristics of various over-current relays: (*a*) definite time, (*b*) IDMT, (*c*) very inverse, and (*d*) extremely inverse.

Earth fault relay

A relay used for the protection of an element of a power systems against earth faults is known as an earth fault relay.

Phase fault relay

A relay used for the protection of an element of a power systems against phase faults is known as an phase fault relay.

Negative sequence relay

A relay for which the actuating quantity is the negative sequence current. When the negative sequence current exceeds a certain value, the relay operates. This type of relay is used to protect electrical machines against overheating due to unbalanced current.

Zero sequence relay

A relay for which the actuating quantity is the zero sequence current. This type of relay is used for earth fault protection.

Reach

This term is mostly used in connection with distance relays. A distance relay operates when the impedance as seen by the relay is less than a preset value. This preset impedance or corresponding distance is called the reach of the relay. In other words, it is the maximum length of the line up to which the relay can protect

Over reach

Some time a relay may operate even when a fault point is beyond its present reach (i.e. its protected length)

Under reach

Sometimes a relay may fail to operate even when the fault point is within its reach, but it is at far end of the protected line. This phenomena is called under reach.

Restricted earth fault protection

This is an English term which may be misunderstood in other countries. It is used in the context of transformer or alternator. It refers to the differential protection of transformers or alternators against ground faults. It is called restricted because its zone of protection is restricted only to the winding of the alternator or transformer. The scheme responds to the faults occurring within its zone of protection. It does not respond to faults beyond its zone of protection.

Unrestricted protection

A protection system which has no clearly defined zone of operation and which achieves selective operation only by time grading.

Percentage distribution of Faults in Power Systems

- Overhead line : 50%
- Underground cables:9%
- Transformer: 10%
- Generators:7%
- Switchgears:12%
- CTs, PTs, other control equipment: 12%

Frequency of occurrence of faults

- L-G : 85%
- L-L :8%
- 2L-G:5%
- 3 phase :2%

Protection classification

1.11 CLASSIFICATION OF PROTECTIVE RELAYS

Protective relays can be classified in various ways depending on the technology used for their construction, their speed of operation, their generation of development, function, etc., and will be discussed in more details in the following chapters.

1.11.1 Classification of Protective Relays Based on Technology

Protective relays can be broadly classified into the following three categories, depending on the technology they use for their construction and operation.

(i) Electromechanical relays

(ii) Static relays

(iii) Numerical relays

Electromechanical Relays

Electromechanical relays are further classified into two categories, i.e., (i) electromagnetic relays, and (ii) thermal relays. *Electromagnetic relays* work on the principle of either electromagnetic attraction or electromagnetic induction. *Thermal relays* utilise the electrothermal effect of the actuating current for their operation.

First of all, electromagnetic relays working on the principle of electromagnetic attraction were developed. These relays were called attracted armature-type electromagnetic relays. This type of relay operates through an armature which is attracted to an electromagnet or through a plunger drawn into a solenoid. Plunger type electromagnetic relays are used for instantaneous units for detecting over current or overvoltage conditions.

Static Relays

Static relays contain electronic circuitry which may include transistors, ICs, diodes and other electronic components. There is a comparator circuit in the relay, which compares two or more currents or voltages and gives an output which is applied to either a slave relay or a thyristor circuit. The slave relay is an electromagnetic relay which finally closes the contact. A static relay containing a slave relay is a semi-static relay. A relay using a thyristor circuit is a wholly static relay. Static relays possess the advantages of having low burden on the CT and VT, fast operation, absence of mechanical inertia and contact trouble, long life and less maintenance. Static relays have proved to be superior to electromechanical relays and they are being used for the protection of important lines, power stations and sub-stations. Yet they have not completely replaced electromechanical relays. Static relays are treated as an addition to the family of relays. Electromechanical relays continue to be in use because of their simplicity and low cost. Their maintenance can be done by less qualified personnel, whereas the maintenance and repair of static relays requires personnel trained in solid state devices.

Numerical Relays

Numerical relays are the latest development in this area. These relays acquire the sequential samples of the ac quantities in numeric (digital) data form through the data acquisition system, and process the data numerically using an algorithm to calculate the fault discriminants and make trip decisions. Numerical relays have been developed because of tremendous advancement in VLSI and computer hardware technology. They are based on numerical (digital) devices, e.g., microprocessors, microcontrollers, Digital Signal Processors (DSPs), etc. At present microprocessor/microcontroller-based numerical relays are widely used. These relays use different relaying algorithms to process the acquired information. Microprocessor/microcontroller-based relays are called numerical relays specifically if they calculate the algorithm numerically. The term 'digital relay' was originally used to designate a previous-generation relay with analog measurement circuits and digital coincidence time measurement (angle measurement) using microprocessors. Now a days the term 'numerical relay' is widely used in place of 'digital relay'. Sometimes, both terms are used in parallel. Similarly, the term 'numerical protection' is widely used in place of 'digital protection'. Sometimes both these terms are also used in parallel.

1.13 CLASSIFICATION OF PROTECTIVE SCHEMES

A protective scheme is used to protect an equipment or a section of the line. It includes one or more relays of the same or different types. The following are the most common protective schemes which are usually used for the protection of a modern power system.

(i) Overcurrent protection

(ii) Distance protection

(iii) Carrier-current protection

(iv) Differential protection

1.13.1 Overcurrent Protection

This scheme of protection is used for the protection of distribution lines, large motors, equipment, etc. It includes one or more overcurrent relays. An overcurrent relay operates when the current exceeds its pick-up value.

1.13.2 Distance Protection

Distance protection is used for the protection of transmission or sub-transmission lines; usually 33 kV, 66 kV and 132 kV lines. It includes a number of distance relays of the same or different types. A distance relay measures the distance between the relay location and the point of fault in terms of impedance, reactance, etc. The relay operates if the point of fault lies within the protected section of the line. There are various kinds of distance relays. The important types are impedance, reactance and mho type. An impedance relay measure the line impedance between the fault point and relay location; a reactance relay measures reactance, and a mho relay measures a component of admittance.

1.13.3 Carrier-Current Protection

This scheme of protection is used for the protection of EHV and UHV lines, generally 132 kV and above. A carrier signal in the range of 50-500 kc/s is generated for the purpose. A transmitter and receiver are installed at each end of a transmission line to be protected. Information regarding the direction of the fault current is transmitted from one end of the line section to the other. Depending on the information, relays placed at each end trip if the fault lies within their protected section. Relays do not trip in case of external faults. The relays are of distance type and their tripping operation is controlled by the carrier signal.

1.13.4 Differential Protection

This scheme of protection is used for the protection of generators, transformers, motors of very large size, bus zones, etc. CTs are placed on both sides of each winding of a machine. The outputs of their secondaries are applied to the relay coils. The relay compares the current entering a machine winding and leaving the same. Under normal conditions or during any external fault, the current entering the winding is equal to the current leaving the winding. But in the case of an internal fault on the winding, these are not equal. This difference in the current actuates the relay. Thus, the relay operates for internal faults and remains inoperative under normal conditions or during external faults. In case of bus zone protection, CTs are placed on the both sides of the bus bar.

1.14 AUTOMATIC RECLOSING

About 90% of faults on overhead lines are of transient nature. Transient faults are caused by lightning or external bodies falling on the lines. Such faults are always associated with arcs. If the line is disconnected from the system for a short time, the arc is extinguished and the fault disappears. Immediately after this, the circuit breaker can be reclosed automatically to restore the supply.

Most faults on EHV lines are caused by lightning. Flashover across insulators takes place due to overvoltages caused by lightning and exists for a short time. Hence, only on instantaneous reclosure is used in the case of EHV lines. There is no need for more than one reclosure for such a situation. For EHV lines, one reclosure in 12 cycles is recommended. A fast reclosure is desired from the stability point of view.

On lines up 33 kV, most faults are caused by external objects such as tree branches, etc. falling on the overhead lines. This is due to the fact that the support height is less than that of the trees. The external objects may not be burnt clear at the first reclosure and may require additional reclosures. Usually three reclosures at 15-120 seconds' intervals are made to clear the fault. Statistical reports show that over 80% faults are cleared after the first reclosure, 10% require the second reclosure and 2% need the third reclosure, while the remaining 8% are permanent faults. If the fault is not cleared after 3 reclosures, it indicates that the fault is of permanent nature. Automatic reclosure are not used on cables as the breakdown of insulation in cables causes a permanent fault.

1.15 CURRENT TRANSFORMERS (CTs) FOR PROTECTION

Current transformers (CTs) are used to obtain reduced current signals from the power systems for the purpose of measurement, control and protection. They reduce the heavy currents of the power system to lower values that are suitable for the operation of relays and other instruments connected to their secondary windings. Besides reducing the current levels, CTs also isolate the relays and instruments circuits from the primary circuit which is a high voltage power circuit and allow the use of standardized current ratings for relays and meters. The current ratings of the secondary windings of the CTs have been standardized so that a degree of interchangeability among relays and meters of different manufacturers can be achieved. Since the standard current ratings of the secondary windings of the CTs are 5 or 1 ampere the protective relays also have the same current rating. The current transformers are designed to withstand fault currents (which may be as high as 50 times the full load current) for a few seconds. Protective relays require reasonably accurate reproduction of the normal and abnormal conditions in the power system for correct sensing and operation. Hence the current transformers should be able to provide current signals to the relays which are faithful reproductions of the primary currents. The measure of a current transformer performance is its ability to accurately reproduce the primary current in secondary.

The requirements of CTs used for relaying are quite different from those of metering (CTs for measuring instruments). CTs used for instrumentation are required to be accurate over the normal working range of currents, whereas CTs used for relaying are required to give a correct ratio up to several times the rated primary current. The CTs used for metering may have very significant errors during fault conditions, when the currents may be several times their normal value for a very short time. Since metering functions are not required during faults, this is not significant. CTs used for relaying are designed to have small errors during fault conditions, whereas their performance during normal steady state condition, when the relay is not required to operate, may not be as accurate.

1.16 VOLTAGE TRANSFORMERS (VTs)

Voltage transformers (VTs) were previously known as potential transformers (PTs). They are used to reduce the power system voltages to lower values and to provide isolation between the high-voltage power network and the relays and other instruments connected to their secondaries. The voltage ratings of the secondary windings of the VTs have been standardized, so that a degree of interchangeability among relays and meters of different manufacturers can be achieved. The secondary windings of the voltage transformers are rated at 110 V line to line. Therefore, the voltage ratings of the voltage (pressure) coils of protective relays and measuring instruments (meters) are also 110 V line to line. The voltage transformers should be able to provide voltage signals to the relays (and meters) which are faithful reproductions of the primary voltages.